

Mental Health Resource Accessibility Tracker

Course: Database Management Systems (CSE370)

1. Executive Summary & Project Vision

The Mental Health Resource Accessibility Tracker is a specialized, database-driven enterprise system designed to bridge the gap between individuals seeking mental health support and healthcare providers, counseling centers, crisis hotlines, and community support groups. Particularly in underserved regions, access to mental health resources suffers from severe information asymmetry, long wait times, inefficient referrals, and resource misallocation. Unlike generic appointment booking tools, this project focuses heavily on automated database logic, real-time analytics, and proactive constraint management. It features intelligent resource matching, crisis escalation triggers, automated shortage alerts, closed-loop referral tracking, and regional underserved zone detection. The system is engineered to serve patients, healthcare providers, NGOs, and healthcare policymakers.

2. Team Division of Labor & Feature Matrix

To ensure equal distribution of technical complexity across all 3 team members, the system is partitioned into three distinct database modules. Basic authentication (Login, Logout, Signup) is excluded from this feature count.

Member Assigned Module	Core Features (DBMS Technical Highlights)
Member 1 Patient Journey & Smart Allocation Engine	1. Smart Resource Matcher (Stored Procedure / Multi-Table Join) 2. Waitlist Priority & Crisis Escalation Engine (Automated Trigger) 3. User Progress & Engagement Tracker (Complex Relational View)
Member 2 Provider Network & Hotline Operations	4. Closed-Loop Referral Lifecycle Tracker (Audit Logic & Foreign Key Constraints) 5. Provider Performance & Reliability Analyzer (Statistical Aggregation) 6. Crisis Hotline Capacity & Status Monitor (Real-Time Database Event)
Member 3 Regional Analytics & Resource Planning	7. Underserved Zone Detector (Complex Analytical Query) 8. Automated Resource Shortage Alerting

	System (Database Trigger) 9. Appointment No-Show & Barrier Pattern Analyzer (Data Aggregation & Grouping)
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3. Detailed Module & Feature Specifications

Module 1: Patient Journey & Smart Allocation Engine (Member 1)

Feature 1: Smart Resource Matcher (Stored Procedure)

Description: Connects a patient seeking help with the most compatible mental health provider or clinic based on multi-parameter relational filtering.

Real-World Example: A patient in Chittagong suffering from PTSD needs a Bengali-speaking therapist with a session fee under 1,500 BDT who accepts insurance. The system queries active providers, calculates geographical proximity and cost constraints, and returns a ranked list of matching specialists.

DBMS Implementation: Stored Procedure parameterized by patient_id, district_id, max_budget, language_code, and specialization_tag. Utilizes multi-table INNER JOIN across Providers, Specializations, Languages, Provider_Languages, and Regions.

Feature 2: Waitlist Priority & Crisis Escalation Engine (Automated Trigger)

Description: Automatically monitors patients waiting for appointments and escalates their queue priority if their waiting time exceeds safety thresholds or if their self-reported crisis level increases.

Real-World Example: If a high-anxiety patient has been on a facility waitlist for over 14 days without an assigned slot, or updates their crisis assessment score from 'Moderate' to 'Severe', the database automatically flags them as 'High Priority' and shifts them ahead in the queue.

DBMS Implementation: Database BEFORE/AFTER UPDATE or INSERT Trigger on Waitlist table. Calculates DATEDIFF(CURRENT_DATE, request_date) and evaluates severity flags to automatically update priority_level = 'CRITICAL' where status = 'Active'.

Feature 3: User Progress & Engagement Tracker (Complex Relational View)

Description: Provides a holistic, chronological timeline of a patient's interaction with the mental health ecosystem, consolidating appointments, accessed resources, therapy completions, and active referrals.

Real-World Example: A psychiatrist viewing a patient's history can see a unified dashboard showing 5 attended counseling sessions, 2 crisis hotline contacts, and 1 completed referral to a psychiatric clinic.

DBMS Implementation: SQL CREATE VIEW aggregating data from Appointments, Referrals, and Resource_Access_Logs using UNION ALL and temporal ordering.

Module 2: Provider Network & Hotline Operations (Member 2)

Feature 4: Closed-Loop Referral Lifecycle Tracker (Audit Logic)

Description: Tracks patient transfers between general therapists, crisis centers, and specialized psychiatric hospitals, verifying whether referrals are fulfilled or dropped.

Real-World Example: A community counselor refers a depressed patient to a clinical psychiatrist. The system logs the referral ID as 'Pending'. Once the patient attends the intake appointment at the psychiatrist's clinic, the database automatically marks the referral as 'Completed'.

DBMS Implementation: Foreign Key constraints connecting Referrals to Appointments and Providers. Uses an AFTER UPDATE trigger on Appointments to set Referral_Status = 'Fulfilled' when the appointment status changes to 'Completed'.

Feature 5: Provider Performance & Reliability Analyzer (Statistical Aggregation)

Description: Aggregates metrics on healthcare providers including patient ratings, session completion rates, response promptness, and cancellation histories.

Real-World Example: The system aggregates quarterly feedback for a therapy clinic. If a provider's completion rate drops below 75% or negative feedback spikes, the database flags the provider profile for administrative performance review.

DBMS Implementation: Analytical query using aggregate functions (AVG(), COUNT(), SUM()) combined with GROUP BY provider_id and HAVING AVG(rating) < 3.5 OR completion_rate < 0.75.

Feature 6: Crisis Hotline Capacity & Status Monitor (Real-Time Database Event)

Description: Tracks the active operational status, call line saturation, and multilingual staffing of crisis hotlines in real time.

Real-World Example: During peak night hours (12 AM - 4 AM), if a suicide prevention hotline reaches 100% line capacity, the database automatically routes new incoming queries to secondary operational hotlines with available capacity.

DBMS Implementation: Scheduled DB event / conditional query checking active call connections against max_capacity in the Hotlines table, updating status flags dynamically.

Module 3: Regional Analytics & Resource Planning (Member 3)

Feature 7: Underserved Zone Detector (Complex Analytical Query)

Description: Analyzes regional demographics against available mental health resources to identify geographical zones with severe shortages of care.

Real-World Example: The database compares the population density of Sylhet Division (e.g., 500,000 residents) against the total active therapists (e.g., 2 therapists), calculating a critical ratio of 1:250,000 and flagging the district as a 'Red Zone' for NGO intervention.

DBMS Implementation: Complex query utilizing Common Table Expressions (WITH clause), window functions (DENSE_RANK(), RATIO_TO_REPORT() or division logic), joining Regions with grouped Providers count by district_id.

Feature 8: Automated Resource Shortage Alerting System (Database Trigger)

Description: Detects facilities or specialized resources that are consistently fully booked over extended periods and generates capacity expansion alerts.

Real-World Example: A child psychology clinic operates at 100% booking capacity for 3 consecutive weeks. A database trigger automatically generates an entry in the System_Alerts table notifying health authorities to allocate more personnel.

DBMS Implementation: Trigger on Appointments table that evaluates rolling 30-day slot utilization ratios. If $\text{booked_slots} / \text{total_capacity} \geq 0.95$, inserts a alert row into System_Alerts.

Feature 9: Appointment No-Show & Barrier Pattern Analyzer (Data Aggregation)

Description: Analyzes missed or canceled appointments across regions, financial tiers, and travel distances to identify structural access barriers.

Real-World Example: Data analysis reveals that 40% of missed appointments in rural districts correlate with travel distances greater than 25 km and low-income categories, providing empirical proof that distance and transportation costs are primary access barriers.

DBMS Implementation: SQL query using COUNT(CASE WHEN status='No-Show' THEN 1 END), grouped by distance_bracket, income_tier, and region_id.

4. Relational Database Schema Overview

The core entity-relationship model consists of the following primary tables and relational links:

- **Users / Patients:** user_id (PK), name, email, phone, address, district_id (FK), income_bracket, preferred_language, created_at, latitude, longitude
- **Providers:** provider_id (PK), name, type (Therapist/Clinic), district_id (FK), session_fee, max_capacity, rating_avg, latitude, longitude, accepts_insurance (Boolean)
- **Hotlines:** hotline_id (PK), provider_id (FK), max_capacity, active_connections, status
- **Languages:** language_code (PK), language_name
- **Provider Languages:** provider_id (FK), language_code (FK) (Composite PK)
- **Specializations:** spec_id (PK), spec_name (Anxiety/PTSD/Depression/etc.)
- **Provider Specializations:** provider_id (FK), spec_id (FK) (Composite PK)
- **Appointments:** appointment_id (PK), patient_id (FK), provider_id (FK), referral_id (FK), appointment_date, status (Completed/Cancelled/No-Show)
- **Waitlist:** waitlist_id (PK), patient_id (FK), spec_id (FK), request_date, crisis_score, priority_level, status

- **Referrals:** referral_id (PK), patient_id (FK), source_provider_id (FK), target_provider_id (FK), referral_date, status (Pending/Fulfilled/Expired)
- **Resource_Access_Logs:** log_id (PK), patient_id (FK), resource_type, access_time
- **Regions:** district_id (PK), district_name, population, risk_index
- **System_Alerts:** alert_id (PK), district_id (FK), alert_type, created_at, status